# Introductory Calculus

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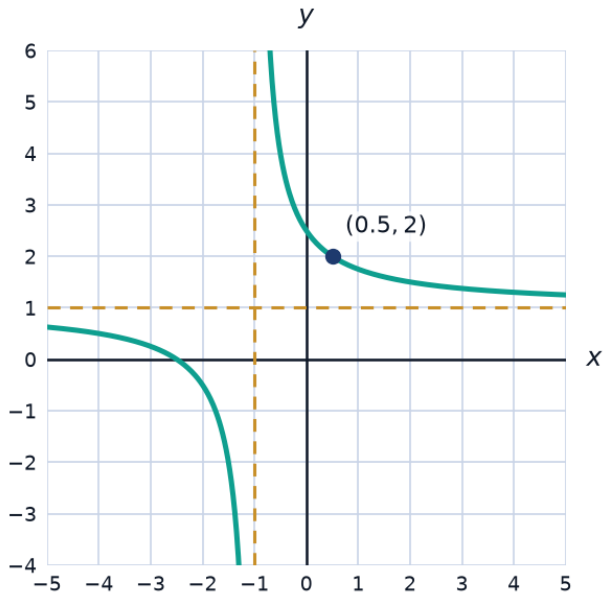
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Direct substitution**

Find $\lim_{x \to 2} (3x^2 - 2x + 1)$.

**2** **[Foundational]** • **Target: Direct substitution**

Find $\lim_{x \to -1} (x^3 - 5x)$.




**3** **[Foundational]** • **Target: Indeterminate forms: factor and cancel**

Find $\lim_{x \to 5} \frac{x^2 - 25}{x - 5}$.

**4** **[Foundational]** • **Target: Indeterminate forms: factor and cancel**

Find $\lim_{x \to -3} \frac{x^2 - 9}{x + 3}$.

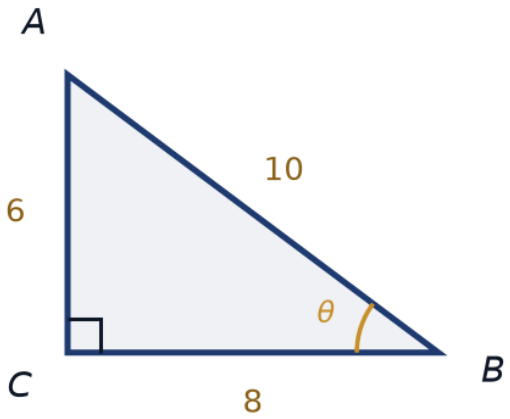
**5** **[Foundational]** • **Target: The derivative as a limit**

Write the difference-quotient definition of $f'(x)$.

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**6** [Foundational] • Target: The derivative as a limit

Using Theorem 5.1 ($f(x) = x^2 \Rightarrow f'(x) = 2x$), find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.




**7** [Foundational] • Target: The derivative as a limit

Using Theorem 5.1, find the slope of the tangent line to $f(x) = x^2$ at $x = -5$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Differential equations

Show that $y = x^2 - 7$ solves $\frac{dy}{dx} = 2x$.

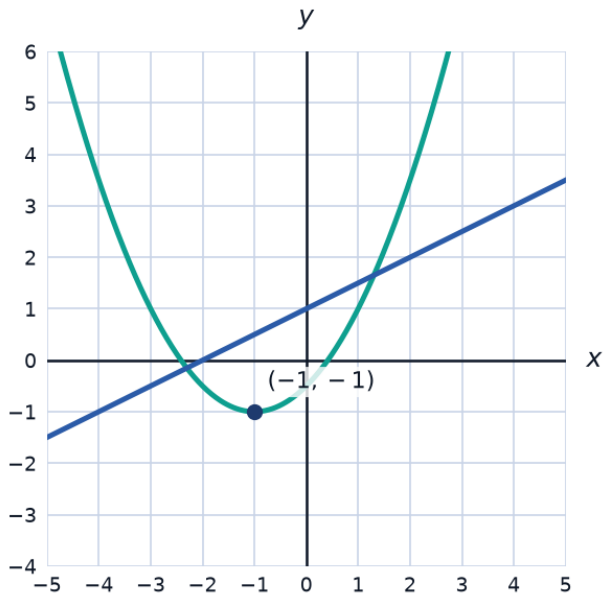
**9** [Intermediate] • Target: Differential equations

Explain why solving a differential equation gives a family of functions rather than a single number.

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**10** [Intermediate] • Target: Indeterminate forms: factor and cancel

Does $y = 5x^2 - 8$ solve $\frac{dy}{dx} = 10x$?




**11** [Intermediate] • Target: The derivative as a limit

Does $y = x^2 + 9$ solve $\frac{dy}{dx} = 2x$?

**12** [Intermediate] • Target: Differential equations

Find $A^{-1}$ for $A = [3, 6;\ -4,\ -4]$.

**13** [Intermediate] • Target: Limits & Continuity

Find $A^{-1}$ for $A = [1, 3;\ -1, 5]$.

**14** [Intermediate] • Target: Derivatives & Power/Product Rules

Find $\lim_{x \to -2} \frac{x^2 - 2x - 8}{x + 2}$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

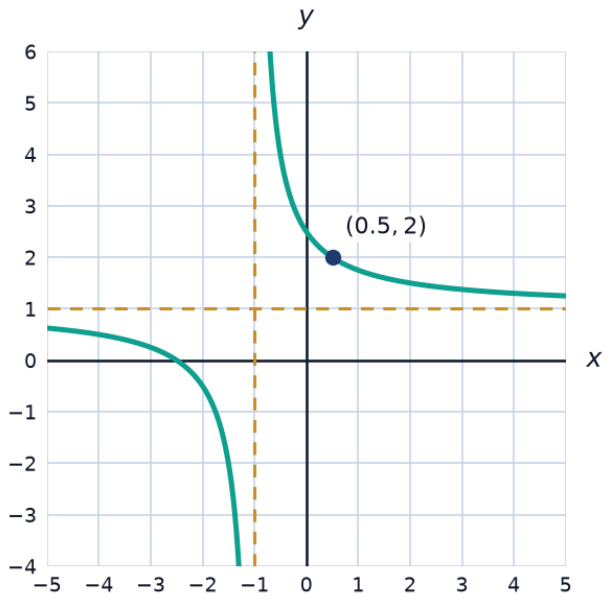
**15** [Challenging] • Target: Chain Rule & Extrema • Cross-Lesson: 4.01 Matrices & Vectors

Find $\lim_{x \to -3} \frac{x^2 + x - 6}{x + 3}$.

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**16** [Challenging] • Target: Definite Integrals & Fundamental Theorem • Cross-Lesson: 6.01 Probability

Find $\lim_{x \to -3} \frac{x^2 + 9x + 18}{x + 3}$.




**17** [Challenging] • Target: Direct substitution • Cross-Lesson: 1.01 Complex Numbers

Find $\lim_{x \to 1} \frac{x^2 + 6x - 7}{x - 1}$.

**18** [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 2.01 Sequences & Series

Find $\lim_{x \to 0} \frac{x^2 + 8x}{x}$.

**19** [Challenging] • Target: The derivative as a limit • Cross-Lesson: 3.01 Functions & Graphs

Given $f(x) = -x^2 + 5x - 4$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(2)$.

**20** [Challenging] • Target: Differential equations • Cross-Lesson: 4.01 Matrices & Vectors

Given $f(x) = 4x^2 + 5x + 9$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(-6)$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Direct substitution

Find $\lim_{x \to 0} (-2x^2 - 6x - 6)$.

**a.** $-6$

**b.** $6$

**c.** $-10$

**d.** $-2$

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**22** [Foundational] • Target: Indeterminate forms: factor and cancel

Find $\lim_{x \to -5} (-3x^2 - 6x + 2)$.

**a.** $-43$

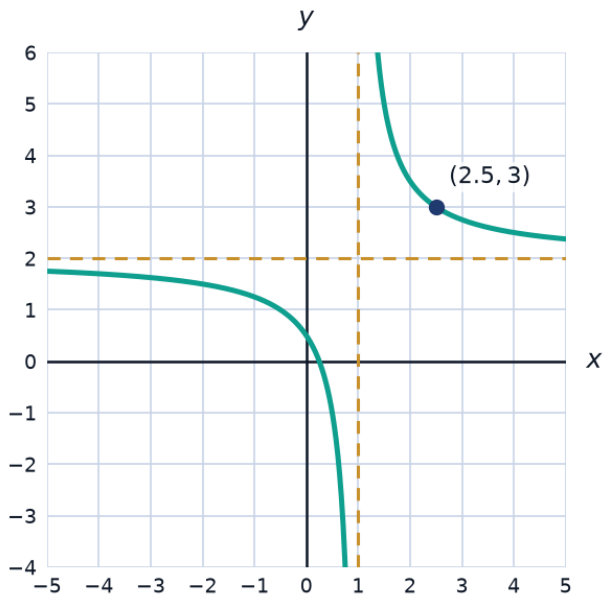
**b.** $-39$

**c.** $-47$

**d.** 43

**23** [Foundational] • Target: The derivative as a limit

Find $\lim_{x \to -3} (x^2 - 4x + 5)$.




**a.** 22

**b.** 26

**c.** $-26$

**d.** 30

**24** [Foundational] • Target: Differential equations

Find $\lim_{x \to -3} (2x^2 + 5x - 1)$.

**a.** $-2$

**b.** 6

**c.** 8

**d.** 2

**25** [Foundational] • Target: Limits & Continuity

Find $\lim_{x \to 4} (2x^2 + 5x + 9)$.

**a.** 57

**b.** $-61$

**c.** 65

**d.** 61

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**26** [Foundational] • Target: Derivatives & Power/Product Rules

Find $\lim_{x \to -5} (-2x^2 - 8x + 4)$.

**a.** 6

**b.** $-10$

**c.** $-6$

**d.** $-2$

**27** [Foundational] • Target: Chain Rule & Extrema

Find $\lim_{x \to -2} (-2x^2 + 5x - 9)$.

**a.** 27

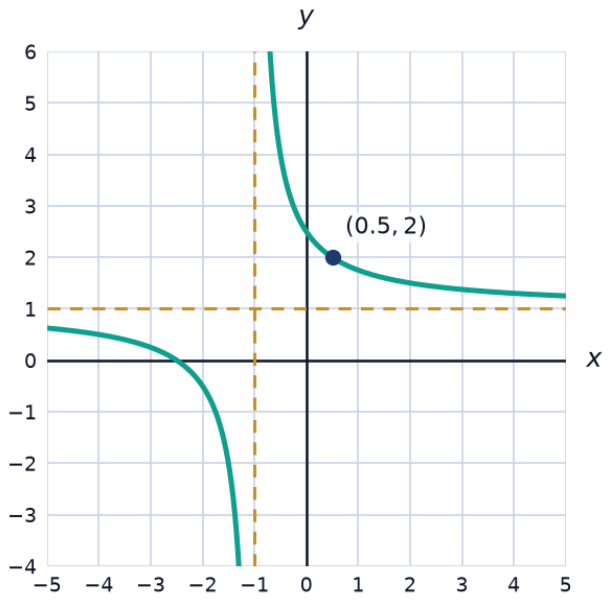
**b.** $-23$

**c.** $-27$

**d.** $-31$

**28** [Foundational] • Target: Definite Integrals & Fundamental Theorem

Find $\lim_{x \to -7} \frac{x^2 + 15x + 56}{x + 7}$.




**a.** 1

**b.** $-2$

**c.** $-1$

**d.** 4

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** **[Intermediate]** • **Target: Direct substitution**

Find $\lim_{x \to -5} \frac{x^2 + 6x + 5}{x + 5}$.

**a.** $-1$

**b.** $-7$

**c.** $-4$

**d.** $4$

**30** **[Intermediate]** • **Target: Indeterminate forms: factor and cancel**

Find $\lim_{x \to 5} \frac{x^2 - 11x + 30}{x - 5}$.

**a.** $1$

**b.** $-1$

**c.** $2$

**d.** $-4$

**31** **[Intermediate]** • **Target: The derivative as a limit**

Find $\lim_{x \to 6} \frac{x^2 - 14x + 48}{x - 6}$.

**a.** $-5$

**b.** $2$

**c.** $-2$

**d.** $1$

**32** **[Intermediate]** • **Target: Differential equations**

Find $\lim_{x \to 7} \frac{x^2 - 4x - 21}{x - 7}$.

**a.** $13$

**b.** $-10$

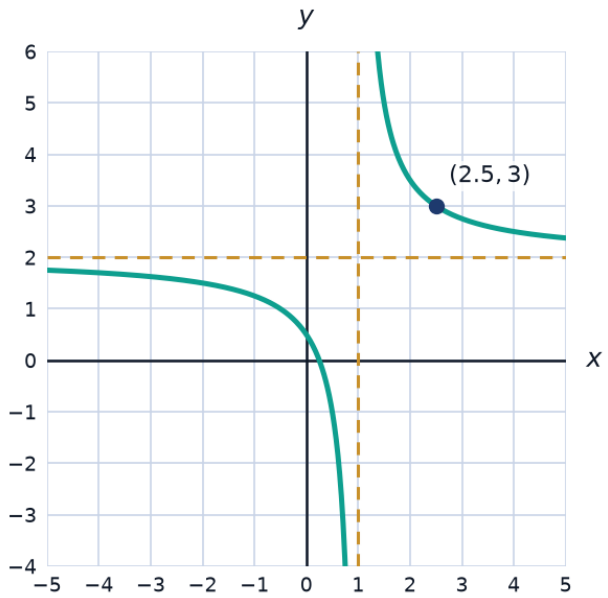
**c.** $7$

**d.** $10$

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**33** [Intermediate] • Target: Limits & Continuity

Find $\lim_{x \to -3} \frac{x^2 + x - 6}{x + 3}$.




**a.** $-2$

**b.** $-8$

**c.** $-5$

**d.** $5$

**34** [Intermediate] • Target: Derivatives & Power/Product Rules

Find $\lim_{x \to 1} \frac{x^2 + 6x - 7}{x - 1}$.

**a.** $5$

**b.** $-8$

**c.** $8$

**d.** $11$

**35** [Intermediate] • Target: Chain Rule & Extrema

Given $f(x) = -3x^2 + 9x - 2$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(3)$.

**a.** $-7$

**b.** $-11$

**c.** $9$

**d.** $-9$

**36** [Intermediate] • Target: Definite Integrals & Fundamental Theorem

Find $\lim_{x \to \infty} \frac{6x + 3}{x - 9}$.

**a.** $\frac{13}{2}$

**b.** $-6$

**c.** $\frac{11}{2}$

**d.** $6$

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**37** [Intermediate] • Target: Direct substitution

Find $\lim_{x\to\infty} \frac{x+5}{3x-8}$.

**a.** $\frac{5}{6}$

**b.** $\frac{1}{3}$

**c.** $-\frac{1}{3}$

**d.** $-\frac{1}{6}$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 2.01 Sequences & Series

Find $\lim_{x\to\infty} \frac{-4x-2}{-4x+4}$.

**a.** $\frac{3}{2}$

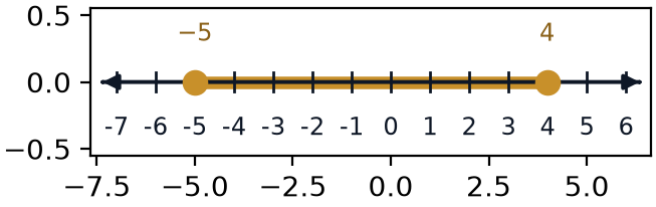
**b.** $1$

**c.** $\frac{1}{2}$

**d.** $-1$

**39** [Challenging] • Target: The derivative as a limit • Cross-Lesson: 3.01 Functions & Graphs

Find the average rate of change of $f(x) = x^2$ on the interval from $x = 3$ to $x = 1$.




**a.** $2$

**b.** $6$

**c.** $-4$

**d.** $4$

**40** [Challenging] • Target: Differential equations • Cross-Lesson: 4.01 Matrices & Vectors

Find the average rate of change of $f(x) = x^2$ on the interval from $x = 3$ to $x = 4$.

**a.** $5$

**b.** $-7$

**c.** $9$

**d.** $7$

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**41** [Challenging] • Target: Limits & Continuity • Cross-Lesson: 6.01 Probability

Evaluate $\lim_{x \to 0} \frac{\tan(3x)}{x}$.

**a.** 3

**b.** 1

**c.** $\frac{1}{3}$

**d.** 0

**42** [Challenging] • Target: Derivatives & Power/Product Rules • Cross-Lesson: 1.01 Complex Numbers

Find the derivative of $f(x) = 4x^3 - 5x^2 + 7x - 9$.

**a.** $12x^2 - 10x + 7$

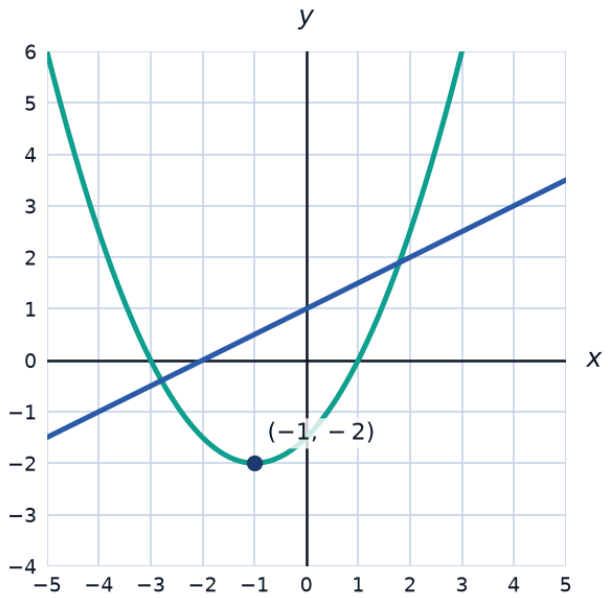
**b.** $12x^2 - 5x + 7$

**c.** $4x^2 - 10x + 7$

**d.** $12x^3 - 10x^2 + 7$

**43** [Challenging] • Target: Derivatives & Power/Product Rules • Cross-Lesson: 2.01 Sequences & Series

What is the derivative of $g(x) = \frac{2x+1}{x-3}$?




**a.** $-\frac{7}{(x-3)^2}$

**b.** $\frac{7}{(x-3)^2}$

**c.** $\frac{2}{(x-3)^2}$

**d.** $-\frac{5}{(x-3)^2}$

**44** [Challenging] • Target: Chain Rule & Extrema • Cross-Lesson: 3.01 Functions & Graphs

At what value of $x$ does $f(x) = 2x^3 - 9x^2 + 12x + 1$ have a local minimum?

**a.** $x = 2$

**b.** $x = 1$

**c.** $x = 0$

**d.** $x = 3$

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**45** [Challenging] • Target: Definite Integrals & Fundamental Theorem • Cross-Lesson: 4.01 Matrices & Vectors

Evaluate the definite integral: $\int_0^3 (3x^2 - 2x + 1)dx$.

**a.** 21

**b.** 18

**c.** 24

**d.** 27

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